Volume Iv Reference

Contents

Mathematical Spaces

0.1 Spaces as Structured Arenas

Definition 0.1.1 (Space)**.** A **space** is a nonempty set.

Proofs

Metric Spaces

0.2 Metric Spaces

0.3 Definitions

Definition 0.3.1 (Metric Space)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

is called a *metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$ *(positive definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

A *metric space* is a pair (X, d), where X is a nonempty set and d is a metric on X.

Definition 0.3.2 (Pseudo-Metric)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

is called a *pseudo-metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, x) = 0$ *(positive semi-definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

Lemma 0.3.3 (Metrics are Pseudo-Metrics)**.** *Every metric on a nonempty set X is a pseudo-metric on X. The converse is false: there exist pseudo-metrics that are not metrics.*

Go to proof.

0.4 Metrics

0.4.1 Norms Induce Metrics

Definition 0.4.1 (Norm)**.** Let V be a real vector space. A *norm* on V is a function

$$\| \cdot \| : V \to \mathbb{R}_{\geq 0}$$

such that for all $x, y \in V$ and all $\lambda \in \mathbb{R}$,

1. $\|x\| = 0$ if and only if $x = 0$ *(positive definiteness)*;

2. $\|\lambda x\| = |\lambda| \, \|x\|$ *(absolute homogeneity)*;

3. $\|x + y\| \le \|x\| + \|y\|$ *(triangle inequality)*.

Theorem 0.4.2 (A Norm Induces a Metric)**.** *Let V be a real vector space and let $\|\cdot\|$ be a norm on V. Define*

$$d_{\|\cdot\|} : V \times V \to \mathbb{R}_{\ge 0}, \qquad d_{\|\cdot\|}(x,y) := \|x - y\|.$$

Then $d_{\|\cdot\|}$ is a metric on V.

Go to proof.

Definition 0.4.3 (Discrete Metric)**.** Let X be a nonempty set. The *discrete metric* on X is the function

$$d_{\text{disc}} : X \times X \to \mathbb{R}_{\ge 0}$$

defined by

$$d_{\text{disc}}(x,y) = \begin{cases} 0, & x = y, \\ 1, & x \ne y. \end{cases}$$

Proposition 0.4.4 (The Discrete Distance is a Metric)**.** *For every nonempty set X, the discrete distance d_{disc} is a metric on X.*

Go to proof.

Definition 0.4.5 (Euclidean Distance on $\mathbb{R}$)**.** The *Euclidean distance* on $\mathbb{R}$ is the function

$$d_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\ge 0}, \qquad d_{\mathbb{R}}(x,y) = |x - y|.$$

Proposition 0.4.6 (The Real-Line Euclidean Distance is a Metric)**.** *The function $d_{\mathbb{R}}(x,y) = |x-y|$ is a metric on $\mathbb{R}$.*

Go to proof.

Definition 0.4.7 (Euclidean Space)**.** For $n \in \mathbb{N}$, the *n-dimensional Euclidean space* is $\mathbb{R}^n$ equipped with its standard coordinate structure, dot product, Euclidean norm, and Euclidean distance.

Definition 0.4.8 (Euclidean Distance on $\mathbb{R}^n$)**.** For $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathbb{R}^n$, the *Euclidean distance* is the function

$$d_2 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\ge 0}, \qquad d_2(x,y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

Theorem 0.4.9 (Cauchy–Schwarz Inequality)**.** *For every $x, y \in \mathbb{R}^n$,*

$$|x \cdot y| \le |x| \, |y|.$$

Equality holds if and only if x and y are linearly dependent.

Go to proof.

Corollary 0.4.10 (Minkowski's Inequality)**.** *For every $x, y \in \mathbb{R}^n$,*

$$|x + y| \le |x| + |y|.$$

Go to proof.

Proposition 0.4.11 (The Euclidean Distance on $\mathbb{R}^n$ is a Metric)**.** *The function d_2 is a metric on $\mathbb{R}^n$.*

Go to proof.

Corollary 0.4.12 (Euclidean Metric)**.** *For every $x, y \in \mathbb{R}^n$, define*

$$d_2(x,y) := \sqrt{|x_1 - y_1|^2 + \cdots + |x_n - y_n|^2}.$$

Then $(\mathbb{R}^n, d_2)$ is a metric space.

Go to proof.

Definition 0.4.13 (Taxicab Metric)**.** For $x, y \in \mathbb{R}^n$, the *taxicab metric* is the function

$$d_1(x,y) := \sum_{i=1}^{n} |x_i - y_i|.$$

Proposition 0.4.14 (Taxicab Metric)**.** *The function d_1 is a metric on $\mathbb{R}^n$.*

Go to proof.

Definition 0.4.15 (p-Metric on $\mathbb{R}^n$)**.** Let $1 \le p < \infty$. For $x, y \in \mathbb{R}^n$, define

$$d_p(x,y) := \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}.$$

Theorem 0.4.16 (d_p is a Metric)**.** *For every $1 \le p < \infty$, the function d_p is a metric on $\mathbb{R}^n$.*

Go to proof.

Definition 0.4.17 (Complex Metric)**.** Identify $\mathbb{C}$ with $\mathbb{R}^2$. The *complex metric* is

$$d_{\mathbb{C}}(z,w) := |z - w|.$$

Proposition 0.4.18 (The Complex Plane is a Metric Space)**.** *The pair $(\mathbb{C}, d_{\mathbb{C}})$ is a metric space.*

Go to proof.

Definition 0.4.19 (Supremum Metric on Bounded Real Sequences)**.** Let $\ell^\infty(\mathbb{R})$ denote the set of all bounded real sequences:

$$\ell^\infty(\mathbb{R}) := \{x = (x_n)_{n\in\mathbb{N}} : \exists M \ge 0 \ \forall n \in \mathbb{N} \ |x_n| \le M\}.$$

For $x, y \in \ell^\infty(\mathbb{R})$, define

$$d_\infty(x,y) := \sup_{n\in\mathbb{N}} |x_n - y_n|.$$

Proposition 0.4.20 (The Supremum Distance on Bounded Sequences is a Metric)**.** *The function d_∞ is a metric on $\ell^\infty(\mathbb{R})$.*

Go to proof.

0.4.2 Point Functions and Pointlike Functions

Definition 0.4.21 (Point Function)**.** Let (X, d) be a metric space and let $z \in X$. The *point function at z* is the function

$$\delta_z : X \to \mathbb{R}_{\ge 0}, \qquad \delta_z(x) := d(z,x).$$

The set of all point functions on X is denoted by

$$\delta(X) := \{\, \delta_z : z \in X \,\}.$$

Theorem 0.4.22 (Point Functions Identify Points)**.** *Let (X, d) be a metric space. The map*

$$X \to \delta(X), \qquad z \mapsto \delta_z$$

is a bijection.

Go to proof.

Theorem 0.4.23 (Point Function Inequalities)**.** *Let (X, d) be a nonempty metric space and let $z \in X$. Then, for all $a, b \in X$,*

$$\delta_z(b) - \delta_z(a) \le d(a,b) \le \delta_z(b) + \delta_z(a),$$

and

$$\delta_z(z) = 0.$$

Go to proof.

Definition 0.4.24 (Pointlike Function)**.** Let (X, d) be a nonempty metric space. A function

$$u : X \to \mathbb{R}_{\geq 0}$$

is *pointlike* on X if, for all $a, b \in X$,

$$u(a) - u(b) \leq d(a, b) \leq u(a) + u(b).$$

Theorem 0.4.25 (Pointlike Functions with a Zero Are Point Functions)**.** *Let (X, d) be a metric space and let $u : X \to \mathbb{R}_{\geq 0}$. Then $u \in \delta(X)$ if and only if u is pointlike on X and*

$$0 \in u(X).$$

In that case, there is a unique $w \in X$ such that $u(w) = 0$, and

$$u = \delta_w.$$

Go to proof.

0.5 Examples of Metric Spaces

Definition 0.5.1 (Discrete Metric Space)**.** Let X be a nonempty set. The *discrete metric space on X* is the metric space

$$(X, d_{\text{disc}}),$$

where d_{disc} is the discrete metric on X.

Definition 0.5.2 (Euclidean Metric Space)**.** For $n \in \mathbb{N}$, the *standard Euclidean metric space* is

$$(\mathbb{R}^n, d_2),$$

where d_2 is the Euclidean metric on $\mathbb{R}^n$.

Definition 0.5.3 (Taxicab Metric Space)**.** For $n \in \mathbb{N}$, the *taxicab metric space* is

$$(\mathbb{R}^n, d_1),$$

where d_1 is the taxicab metric on $\mathbb{R}^n$.

Definition 0.5.4 (p-Metric Space)**.** Let $1 \leq p < \infty$ and let $n \in \mathbb{N}$. The *p-metric space* is

$$(\mathbb{R}^n, d_p),$$

where d_p is the p-metric on $\mathbb{R}^n$.

Definition 0.5.5 (Complex Metric Space)**.** The *complex metric space* is

$$(\mathbb{C}, d_{\mathbb{C}}),$$

where $d_{\mathbb{C}}$ is the usual Euclidean distance on the complex plane.

Definition 0.5.6 (Bounded Sequence Metric Space)**.** The *bounded sequence metric space* is

$$(\ell^\infty(\mathbb{R}), d_\infty),$$

where $\ell^\infty(\mathbb{R})$ is the set of bounded real sequences and d_∞ is the supremum metric.

0.6 Distances, Boundedness, and Balls

0.6.1 Distances and Boundedness

Definition 0.6.1 (Diameter)**.** Let (X, d) be a metric space and let $A \subseteq X$ be nonempty. The *diameter* of A is

$$\operatorname{diam}(A) := \sup\{\, d(a, b) : a, b \in A \,\},$$

whenever this supremum is finite, and is $+\infty$ otherwise.

Definition 0.6.2 (Distance from a Point to a Set)**.** Let (X, d) be a metric space, let $x \in X$, and let $A \subseteq X$ be nonempty. The *distance from x to A* is

$$d(x, A) := \inf\{\, d(x, a) : a \in A \,\}.$$

Definition 0.6.3 (Distance Between Sets)**.** Let (X, d) be a metric space, and let $A, B \subseteq X$ be nonempty. The *distance between A and B* is

$$d(A, B) := \inf\{\, d(a, b) : a \in A,\ b \in B \,\}.$$

Theorem 0.6.4 (Distance to a Union)**.** *Let (X, d) be a metric space, let $x \in X$, and let $A, B \subseteq X$ be nonempty. Then*

$$d(x, A \cup B) = \min\{d(x, A), d(x, B)\}.$$

Go to proof.

Theorem 0.6.5 (Distance to an Intersection Bound)**.** *Let (X, d) be a metric space, let $x \in X$, and let $A, B \subseteq X$ have nonempty intersection. Then*

$$d(x, A \cap B) \geq \max\{d(x, A), d(x, B)\}.$$

Go to proof.

Definition 0.6.6 (Metric Bounded Set)**.** Let (X, d) be a metric space and let $A \subseteq X$. The set A is *bounded* if there exist $x \in X$ and $R > 0$ such that

$$d(x, a) \leq R$$

for every $a \in A$.

Theorem 0.6.7 (Equivalent Boundedness Criterion)**.** *Let (X, d) be a metric space and let $A \subseteq X$. Then A is bounded if and only if there exist $x \in X$ and $R > 0$ such that*

$$d(x, a) < R$$

for every $a \in A$.

Theorem 0.6.8 (Center Independence)**.** *Let (X, d) be a metric space. If $A \subseteq X$ and there exist $x \in X$ and $R > 0$ such that $d(x, a) \leq R$ for every $a \in A$, then for every $y \in X$,*

$$d(y, a) \leq R + d(x, y)$$

for every $a \in A$. Thus boundedness does not depend on the chosen center.

Theorem 0.6.9 (Subsets of Bounded Sets Are Bounded)**.** *Let (X, d) be a metric space. If $A \subseteq B \subseteq X$ and B is bounded, then A is bounded.*

Theorem 0.6.10 (Finite Sets Are Bounded)**.** *Every finite subset of a metric space is bounded.*

Theorem 0.6.11 (Finite Unions of Bounded Sets Are Bounded). *Let (X, d) be a metric space, let $n \in \mathbb{N}$, and let $A_1, \ldots, A_n \subseteq X$. If $A_1, \ldots, A_n$ are bounded, then*

$$\bigcup_{k=1}^{n} A_k$$

is bounded.

Theorem 0.6.12 (Boundedness via Diameter). *Let (X, d) be a metric space and let $A \subseteq X$ be nonempty. Then A is bounded if and only if there exists $C > 0$ such that*

$$d(a, b) \leq C$$

for all $a, b \in A$.

Theorem 0.6.13 (Closure of a Bounded Set Is Bounded). *Let (X, d) be a metric space. If $A \subseteq X$ is bounded, then $\mathrm{cl}(A)$ is bounded.*

Theorem 0.6.14 (Compact Sets Are Bounded). *Every compact subset of a metric space is bounded.*

Theorem 0.6.15 (Totally Bounded Sets Are Bounded). *Every totally bounded subset of a metric space is bounded.*

0.6.2 Balls and Local Neighborhoods

Definition 0.6.16 (Open Ball). Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *open ball* of radius r centered at x is

$$B_d(x; r) := \{\, y \in X : d(y, x) < r \,\}.$$

Definition 0.6.17 (Closed Ball). Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *closed ball* of radius r centered at x is

$$\overline{B}_d(x; r) := \{\, y \in X : d(y, x) \leq r \,\}.$$

Lemma 0.6.18 (Key Local Ball Lemma). *Let (X, d) be a metric space. If $y \in B_d(x; r)$, then*

$$B_d(y; r - d(x, y)) \subseteq B_d(x; r).$$

Theorem 0.6.19 (Ball Nesting). *Let (X, d) be a metric space, let $x \in X$, and let $0 < r \leq s$. Then*

$$B_d(x; r) \subseteq B_d(x; s).$$

If $r < s$, then also

$$\overline{B}_d(x; r) \subseteq B_d(x; s).$$

Theorem 0.6.20 (Ball Containment by Center Distance). *Let (X, d) be a metric space, let $x, y \in X$, and let $r, s > 0$.*

If

$$d(x, y) + r \leq s,$$

then

$$B_d(x; r) \subseteq B_d(y; s).$$

If

$$d(x, y) + r < s,$$

then

$$\overline{B}_d(x; r) \subseteq B_d(y; s).$$

Theorem 0.6.21 (Intersecting Balls Imply Center-Distance Bound)**.** *Let (X, d) be a metric space, let $x, y \in X$, and let $r, s > 0$. If*

$$B_d(x; r) \cap B_d(y; s) \neq \varnothing,$$

then

$$d(x, y) < r + s.$$

Theorem 0.6.22 (Closed Balls Contain Open Balls)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r > 0$,*

$$B_d(x; r) \subseteq \overline{B}_d(x; r).$$

0.7 Open and Closed Sets

Definition 0.7.1 (Metric Open Set)**.** Let (X, d) be a metric space. A subset $E \subseteq X$ is *open in* (X, d) if either $E = \varnothing$ or, for every $x \in E$, there exists $\varepsilon > 0$ such that

$$B_d(x; \varepsilon) \subseteq E.$$

Theorem 0.7.2 (Open Balls are Open)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r > 0$, the open ball $B_d(x; r)$ is open in (X, d).*

Go to proof.

Theorem 0.7.3 (Metric Open Set Closure Properties)**.** *Let X be a metric space. Then:*

1. *the empty set $\varnothing$ and the space X are open sets;*
2. *an arbitrary union of open subsets of X is open;*
3. *any finite intersection of open subsets of X is open.*

Go to proof.

Definition 0.7.4 (Metric Closed Set)**.** Let (X, d) be a metric space. A subset $E \subseteq X$ is *closed in* (X, d) if its complement $E^c = X \setminus E$ is open in (X, d).

Theorem 0.7.5 (Complement of a Closed Ball Is Open)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r \geq 0$,*

$$X \setminus \overline{B}_d(x; r) = \{\, y \in X : d(x, y) > r \,\}$$

is open.

Theorem 0.7.6 (Closed Balls Are Closed)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r \geq 0$, the closed ball*

$$\overline{B}_d(x; r) = \{\, y \in X : d(x, y) \leq r \,\}$$

is closed.

Definition 0.7.7 (Metric Closure)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *closure* of E, denoted $\mathrm{cl}(E)$, is

$$\mathrm{cl}(E) = \{x \in X : \forall \varepsilon > 0,\ B_d(x; \varepsilon) \cap E \neq \varnothing\}.$$

Theorem 0.7.8 (Closure of an Open Ball Is Contained in the Closed Ball)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r > 0$,*

$$\mathrm{cl}\big(B_d(x; r)\big) \subseteq \overline{B}_d(x; r).$$

Definition 0.7.9 (Dense Subset)**.** Let (X, d) be a metric space and let $A \subseteq X$. The set A is *dense in* X if

$$\mathrm{cl}(A) = X.$$

Definition 0.7.10 (Metric Interior Point)**.** Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. The point x is an *interior point of E* if there exists $\delta > 0$ such that

$$B_d(x; \delta) \subseteq E.$$

Definition 0.7.11 (Metric Interior)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *interior of E* is the set of all interior points of E. It is denoted by

$$E^\circ.$$

Theorem 0.7.12 (Interior is the Largest Open Subset)**.** *Let E be a subset of a metric space X. Then E° is the largest open subset of X contained in E. In other words:*

1. *$E^\circ \subseteq E$;*
2. *E° is open;*
3. *$E^\circ \supseteq O$, for every open set $O \subseteq E$.*

Go to proof.

Definition 0.7.13 (Metric Boundary)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *boundary* of E is

$$\partial E = \mathrm{cl}(E) \cap \mathrm{cl}(X \setminus E).$$

0.8 Limit Points and Isolated Points

Definition 0.8.1 (Metric Limit Point and Isolated Point)**.** Let E be a subset of a metric space (X, d), and let $x \in X$.

1. The point x is a *limit point of E* if, for every $\varepsilon > 0$, the ball $B_d(x; \varepsilon)$ contains a point of $E \setminus \{x\}$.
2. The point x is an *isolated point of E* if $x \in E$ and x is not a limit point of E.

Theorem 0.8.2 (Closed Sets Contain Their Limit Points)**.** *Let X be a metric space and let $E \subseteq X$. Then E is closed if and only if $E' \subseteq E$.*

Go to proof.

Theorem 0.8.3 (Sequential Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if there exists a sequence of distinct terms in E that converges to x.*

Go to proof.

Corollary 0.8.4 (Neighborhood Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if every neighborhood of x contains infinitely many points of E.*

Go to proof.

Theorem 0.8.5 (Bolzano–Weierstrass for Bounded Infinite Subsets)**.** *Every bounded infinite subset of $\mathbb{R}^m$ has a limit point in $\mathbb{R}^m$.*

Go to proof.

Proposition 0.8.6 (The Derived Set is Closed)**.** *The set of limit points of any set is closed.*

Go to proof.

0.9 Sequential Convergence

Definition 0.9.1 (Metric Sequential Convergence). Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is *convergent in* X if there exists $x_0 \in X$ such that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_0) < \varepsilon \qquad \text{for all } n \geq N.$$

In this case, $\{x_n\}$ converges to x_0, written

$$x_n \to x_0, \qquad x_0 = \lim_{n\to\infty} x_n.$$

Definition 0.9.2 (Metric Neighborhood). Let (X, d) be a metric space and let $x \in X$. A subset $U \subseteq X$ is a *neighborhood of* x if there exists $\delta > 0$ such that

$$B_d(x; \delta) \subseteq U.$$

Definition 0.9.3 (Epsilon Neighborhood). Let (X, d) be a metric space, let $x \in X$, and let $\varepsilon > 0$. The *ε-neighborhood of* x is the open ball

$$B_d(x; \varepsilon) = \{y \in X : d(y, x) < \varepsilon\}.$$

Theorem 0.9.4 (Metric Neighborhood Criterion for Sequential Convergence). *Let (X, d) be a metric space, let $\{x_n\}$ be a sequence in X, and let $x \in X$. Then*

$$x_n \to x$$

if and only if every neighborhood of x contains all but finitely many terms of $\{x_n\}$.

Go to proof.

Definition 0.9.5 (Metric Cauchy Sequence). Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is *Cauchy* if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_m) < \varepsilon \qquad \text{for all } n, m \geq N.$$

Theorem 0.9.6 (Metric Convergent Sequences Have Unique Limits). *In metric spaces, convergent sequences have unique limits.*

Go to proof.

Theorem 0.9.7 (Metric Cauchy Sequence with Convergent Subsequence). *Let $\{x_n\}$ be a Cauchy sequence in a metric space (X, d), let $x \in X$, and let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$ such that*

$$\lim_{k\to\infty} x_{n_k} = x.$$

Then $x_n \to x$.

Go to proof.

Theorem 0.9.8 (Coordinatewise Sequences in $\mathbb{R}^m$). *Let $\{x_n\}$ be a sequence in $\mathbb{R}^m$, and let $x_0 \in \mathbb{R}^m$. Write*

$$x_n = \left(x_n^{(1)}, \ldots, x_n^{(m)}\right) \qquad \textit{for all } n \in \mathbb{N} \cup \{0\}.$$

Then:

1. *$x_n \to x_0$ if and only if $x_n^{(j)} \to x_0^{(j)}$ for every $j = 1, \ldots, m$.*
2. *$\{x_n\}$ is Cauchy if and only if $\{x_n^{(j)}\}$ is Cauchy for every $j = 1, \ldots, m$.*

Go to proof.

Theorem 0.9.9 (Euclidean Cauchy Sequences Converge). *Every Cauchy sequence in $\mathbb{R}^m$ is convergent in $\mathbb{R}^m$.*

Go to proof.

Theorem 0.9.10 (Bolzano–Weierstrass in $\mathbb{R}^m$). *Every bounded sequence in $\mathbb{R}^m$ contains a subsequence that converges in $\mathbb{R}^m$.*

Go to proof.

Proofs

Topology

0.10 Topology

0.11 Definitions

Definition 0.11.1 (Topological Space)**.** Let X be a set. A *topology* on X is a collection $\tau \subseteq \mathcal{P}(X)$ such that:

1. $\emptyset \in \tau$ and $X \in \tau$;
2. if $\{U_i\}_{i \in I}$ is any family of sets in τ, then $\bigcup_{i \in I} U_i \in \tau$;
3. if $U_1, \ldots, U_n \in \tau$, then $\bigcap_{k=1}^{n} U_k \in \tau$.

A *topological space* is a pair (X, τ), where X is a set and τ is a topology on X. The members of τ are called the *open sets* of the space.

Definition 0.11.2 (Discrete Topology)**.** Let X be a set. The *discrete topology* on X is the topology

$$\tau_{\text{disc}} = \mathcal{P}(X).$$

Thus every subset of X is open in the topological space (X, τ_{disc}).

Definition 0.11.3 (Sierpinski Topology)**.** Let $X = \{0, 1\}$. The *Sierpinski topology* on X is the topology

$$\tau_{\text{S}} = \{\emptyset, \{1\}, X\}.$$

The topological space (X, τ_{S}) is called the *Sierpinski space.*

Definition 0.11.4 (Cofinite Topology)**.** Let X be a set. The *cofinite topology* on X is the topology

$$\tau_{\text{cof}} = \{\, U \subseteq X : X \setminus U \text{ is finite} \,\} \cup \{\emptyset\}.$$

Thus a subset of X is open exactly when it is empty or its complement in the ambient set is finite.

Definition 0.11.5 (Trivial Topology)**.** Let X be a set. The *trivial topology* on X is the topology

$$\tau_{\text{triv}} = \{\emptyset, X\}.$$

Thus only $\emptyset$ and X are open in the topological space (X, τ_{triv}).

0.12 Topological Space Subsets

Definition 0.12.1 (Closed Subset of a Topological Space)**.** Let (X, τ) be a topological space, and let $F \subseteq X$. The subset F is *closed in* (X, τ) if its complement

$$X \setminus F$$

is open in (X, τ).

Theorem 0.12.2 (Closed-Set Characterization of Topologies). *Let X be a set, and let $\mathcal{C}$ be a collection of subsets of X. Suppose that:*

1. *$\emptyset \in \mathcal{C}$ and $X \in \mathcal{C}$;*
2. *the intersection of any family of elements of $\mathcal{C}$ belongs to $\mathcal{C}$;*
3. *the union of any two elements of $\mathcal{C}$ belongs to $\mathcal{C}$.*

Then there is a unique topology τ on X whose family of closed sets is exactly $\mathcal{C}$.

Go to proof.

Definition 0.12.3 (Finer and Coarser Topologies). Let τ_1 and τ_2 be topologies on a set X. The topology τ_1 is *finer than* τ_2 if

$$\tau_2 \subseteq \tau_1.$$

Equivalently, τ_2 is *coarser than* τ_1. The two topologies are *comparable* if one is finer than the other.

0.13 Basis

Definition 0.13.1 (Basis for a Topology). Let (X, τ) be a topological space. A collection $\mathcal{B} \subseteq \tau$ is a *basis for the topology* τ if every open set $O \in \tau$ is the union of some subcollection of $\mathcal{B}$. The elements of $\mathcal{B}$ are called *basic elements* or *basic open sets*.

Proposition 0.13.2 (Pointwise Characterization of a Basis). *Let (X, τ) be a topological space, and let $\mathcal{B} \subseteq \tau$. Then $\mathcal{B}$ is a basis for τ if and only if for every open set $O \in \tau$ and every $x \in O$, there is $B \in \mathcal{B}$ such that*

$$x \in B \subseteq O.$$

Consequently, a subset $A \subseteq X$ is open if and only if for each $x \in A$, there is $B \in \mathcal{B}$ such that

$$x \in B \subseteq A.$$

Go to proof.

Proofs

Set Algebra

0.14 Set Algebras

0.15 Systems of Sets

0.15.1 Systems of Sets

Definition 0.15.1 (Semiring of Sets)**.** A collection $\mathcal{S}$ of subsets of a set X is a *semiring* if:

(i) $\varnothing \in \mathcal{S}$;

(ii) $A, B \in \mathcal{S}$ implies $A \cap B \in \mathcal{S}$;

(iii) for any $A, B \in \mathcal{S}$, the difference $A \setminus B$ is a *finite disjoint union* of members of $\mathcal{S}$: there exist pairwise disjoint $C_1, \ldots, C_n \in \mathcal{S}$ with $A \setminus B = C_1 \sqcup \cdots \sqcup C_n$.

Definition 0.15.2 (Ring of Sets)**.** A collection $\mathcal{R}$ of subsets of X is a *ring of sets* if:

(i) $\varnothing \in \mathcal{R}$;

(ii) $A, B \in \mathcal{R}$ implies $A \cup B \in \mathcal{R}$;

(iii) $A, B \in \mathcal{R}$ implies $A \setminus B \in \mathcal{R}$.

Definition 0.15.3 (Algebra of Sets)**.** A collection $\mathcal{A}$ of subsets of X is an *algebra of sets* (or *Boolean algebra of sets*, or *field of sets*) if:

(i) $X \in \mathcal{A}$;

(ii) $A \in \mathcal{A}$ implies $A^c = X \setminus A \in \mathcal{A}$;

(iii) $A, B \in \mathcal{A}$ implies $A \cup B \in \mathcal{A}$.

Definition 0.15.4 (σ-Algebra)**.** A collection $\mathcal{M}$ of subsets of X is a *σ-algebra* (or *σ-field*) on X if:

(i) $X \in \mathcal{M}$;

(ii) $A \in \mathcal{M}$ implies $A^c \in \mathcal{M}$;

(iii) if $\{A_n\}_{n \in \mathbb{N}}$ is a countable family in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is called a *measurable space*.

Definition 0.15.5 (Topology)**.** A *topology* on a set X is a collection τ of subsets of X satisfying:

(i) $\varnothing \in \tau$ and $X \in \tau$;

(ii) if $\{U_\alpha\}_{\alpha \in I}$ is any family of sets in τ (indexed by any set I), then $\bigcup_{\alpha \in I} U_\alpha \in \tau$;

(iii) if $U_1, \dots, U_n \in \tau$ (finitely many), then $U_1 \cap \cdots \cap U_n \in \tau$.

The pair (X, τ) is a *topological space*. Members of τ are called *open sets*.

Theorem 0.15.6 (Intersection of σ-algebras is a σ-algebra)**.** *Let $\{\mathcal{M}_\alpha\}_{\alpha \in I}$ be any family of σ-algebras on X (indexed by any set I). Then*

$$\mathcal{M} \;:=\; \bigcap_{\alpha \in I} \mathcal{M}_\alpha$$

is a σ-algebra on X. *Go to proof.*

Definition 0.15.7 (Generated σ-Algebra)**.** Let $\mathcal{F}$ be any collection of subsets of X. The *σ-algebra generated by $\mathcal{F}$*, written $\sigma(\mathcal{F})$, is the smallest σ-algebra on X that contains $\mathcal{F}$:

$$\sigma(\mathcal{F}) \;:=\; \bigcap \left\{\, \mathcal{M} \;\middle|\; \mathcal{M} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{F} \subseteq \mathcal{M} \,\right\}.$$

The collection $\mathcal{F}$ is called a *generating family* for $\sigma(\mathcal{F})$.

Definition 0.15.8 (Borel σ-Algebra)**.** Let (X, τ) be a topological space. The *Borel σ-algebra* on X, written $\mathcal{B}(X)$, is the σ-algebra generated by the open sets:

$$\mathcal{B}(X) \;:=\; \sigma(\tau).$$

Members of $\mathcal{B}(X)$ are called *Borel sets*.

0.16 The Power Set and Characteristic Functions

0.16.1 The Power Set and Characteristic Functions

Proofs

Algebras of Sets

0.17 Set Operations and Hierarchy

0.18 Boolean Algebra of Sets

0.19 Rings of Sets

0.20 Sigma Algebras

0.21 Borel Sets

Proofs

Measure Spaces

Proofs

Mathematical Spaces